

Zihan Fang

zihan.fang@vanderbilt.edu | 615 853 1277 | Website | LinkedIn

Research Focus

I study human-centered AI in software engineering. I design and evaluate AI-powered code tools that fit developers' workflows, support developers, and are trustworthy in real practice. Using cognitive measures and mixed-methods studies, I examine how developers engage with and perceive these AI tools, and turn those insights into actionable feedback that improves AI support for coding. I also study open-source development and collaboration dynamics. Ultimately, my goal is to build trustworthy, reproducible, and usable AI that enhances developer productivity and fosters effective learning and coding outcomes.

Education

Vanderbilt University , Ph.D. in Computer Science	Aug. 2023 – Present
• Research Focus: Human-AI Interaction for Software Engineering; Open-Source Software Development	
Vanderbilt University , M.S. in Data Science	Aug. 2021 – May. 2023
• GPA: 3.9/4.0	
Soochow University , B.S. in Information Management	Sept. 2016 – Jun. 2020
• GPA: 3.8/4.0	

Publications

- [1] **Zihan Fang**, Yueke Zhang, Thomas Zimmermann, Denae Ford, Yu Huang: *Contribution Patterns in Open Source Software for Social Good: Dynamics, Individuals, and Impact*. In *Proceedings of the 29th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW'2026)*
- [2] Yueke Zhang, **Zihan Fang**, Greg Trafton, Daniel Levin, Kevin Leach, Yu Huang: *CodeACT-R: A Cognitive Simulation Framework for Code Reading*. In *Companion of the 40th IEEE/ACM International Conference on Automated Software Engineering (ASE'2025-NIER)*
- [3] **Zihan Fang**, Jiliang Li, Anda Liang, Gina R. Bai, Yu Huang: *A Comparative Study on ChatGPT and Checklist as Support Tools for Unit Testing Education*. In *Companion of the 33rd ACM International Conference on the Foundations of Software Engineering (FSE'2025-SEET)*
- [4] **Zihan Fang**, Yu Huang: *"Math Is a Pain!": Understanding Challenges and Needs of the Machine Learning Community on Stack Overflow*. In *Proceedings of the 27th ACM Conference on Computer-Supported Cooperative Work and Social Computing (CSCW'2024)*
- [5] **Zihan Fang**, Madeline Endres, Thomas Zimmermann, Denae Ford, Westley Weimer, Kevin Leach, Yu Huang: *A Four-Year Study of Student Contributions to OSS vs. OSS4SG with a Lightweight Intervention*. In *Proceedings of the 31st ACM Foundations of Software Engineering (FSE'2023, SIGSOFT Distinguished Paper Award)*

Paper Under Review

- [1] **Zihan Fang**, Yifan Zhang, Yueke Zhang, Kevin Leach, Yu Huang: *DPO-F+: Aligning Code Repair Feedback with Developer Preferences*.
- [2] Yueke Zhang, Yifan Zhang, **Zihan Fang**, Kevin Leach, Yu Huang: *ScanCoder: Leveraging Human Attention Patterns to Enhance LLMs*.
- [3] **Zihan Fang**, Zachary Karas, Toby Jia-Jun Li, Collin McMillan, Yu Huang: *Investigating the Feasibility of Conducting Webcam-Based Eye-Tracking Studies in Code Comprehension*.
- [4] **Zihan Fang**, Weizhe Jiao, Robert Tairas, Yu Huang: *Novices Comprehending Functional vs. Imperative Programs: An Eye-Tracking Study*.
- [5] Yueke Zhang, **Zihan Fang**, Kevin Leach, Yu Huang: *An Investigation of Subdomain Takeover Attacks in GitHub Pages*.
- [6] **Zihan Fang**, Janice Chung, Ruijia Chen, Yuhang Zhao, Yu Huang: *A Comparative Human Study of Novice Dyslexic Programmers*.

Academic and Industry Experience

Research Assistant, Vanderbilt University Aug 2023 – Present

Teaching Assistant, Vanderbilt University Aug 2022 – May 2024

- DS3100 Fundamentals of Data Science. Fall 22, Spring 23
- CS3265 Database Management Systems, Fall 23
- CS4278/5278 Principles of Software Engineering, Spring 24

Data Scientist, Tencent Inc. Aug 2021 – Dec 2021

- Productionized a streaming identity-verification ML service on Spark/Flink, moving from a static rules engine to a hybrid ML+rules system; achieved **90%** online accuracy with sub-second latency. (Python, TensorFlow, Spark Structured Streaming, SQL)
- Designed and maintained feature/ETL pipelines across semi-structured and structured sources using PySpark+Hive; partitioning, caching, and query tuning cut pipeline runtime by **30%**.
- Orchestrated jobs and data quality checks (Airflow-style scheduling, CI for SQL), enabling reproducible training/eval and faster iteration with product and compliance stakeholders.

Data Analyst Intern, DiDi Inc. Jan 2021 – Jun 2021

- Segmented drivers via PCA/K-means on HiveSQL features and built profiles tied to retention/PCT/cancellation KPIs; targeted recruiting and onboarding cut acquisition cost by **20%**. (HiveSQL, Python)
- Quantified BD-visit ROI with cohort/propensity analyses and operational dashboards (Tableau); optimized cadence and targeting to raise conversion by **10%**.

Awards

- Ignite Grant at Vanderbilt University
- LIVE Horizon Grant at Vanderbilt University
- C. F. Chen Best Paper Award in the Department of Computer Science at Vanderbilt University
- Richard Bennett/Dorothy Danforth Compton Prize Scholarship at Vanderbilt University
- SIGSOFT Distinguished Paper Award at FSE 2023
- NSF Student Travel Award 2023

Service

SIGCSE TS 2025: PC member (Posters)

SIGCSE TS 2025: PC member (Computing Education Research)